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- **Problem Statement ID – SIH26262**
- **Problem Statement Title- Integrated AICTE Approval Workflow Automation, Compilance Monitoring and Alert Management System**
- **Theme- Smart Automation**
- **PS Category- Software**
- **Team ID-198190**
- **Team Name (Registered on portal)- Uncut Gems**

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VERIFLOW

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Problem:

10,000+ technical institutions face fragmented manual tracking across Scrutiny, EVC, and SAC/SHC, causing application backlogs, delayed inspections, and undetected non-compliance (e.g., **APH Clause 2.1(e) Live Lease rule**).

Our Idea :

A unified, handbook-aware digital orchestration engine featuring deterministic state machines, real-time pendency tracking, and proactive document compliance validation.

Proposed Solution :

- **Dynamic APH Routing:** Automated path assignment across Scrutiny, EVC, & SAC/SHC.
- **Intelligent Document Parsing:** Extracts dates and metadata to flag statutory defects.
- **Proactive SLA Tracking:** Stage-wise dwell time alerts with hierarchical escalation.
- **Unified Command Center:** Role-based visibility for institutions, officers, and committees.

Innovation/Uniqueness:

- **Handbook-Trained State Machine:** Enforces deterministic regulatory logic.
- **Live Lease Compliance Guard:** Real-time expiry and duration validation.
- **Clause-Linked Evidence Auditing:** Connects extracted data directly to APH clauses.
- **Human-in-the-Loop Safeguards:** Automates triage while preserving statutory discretion.

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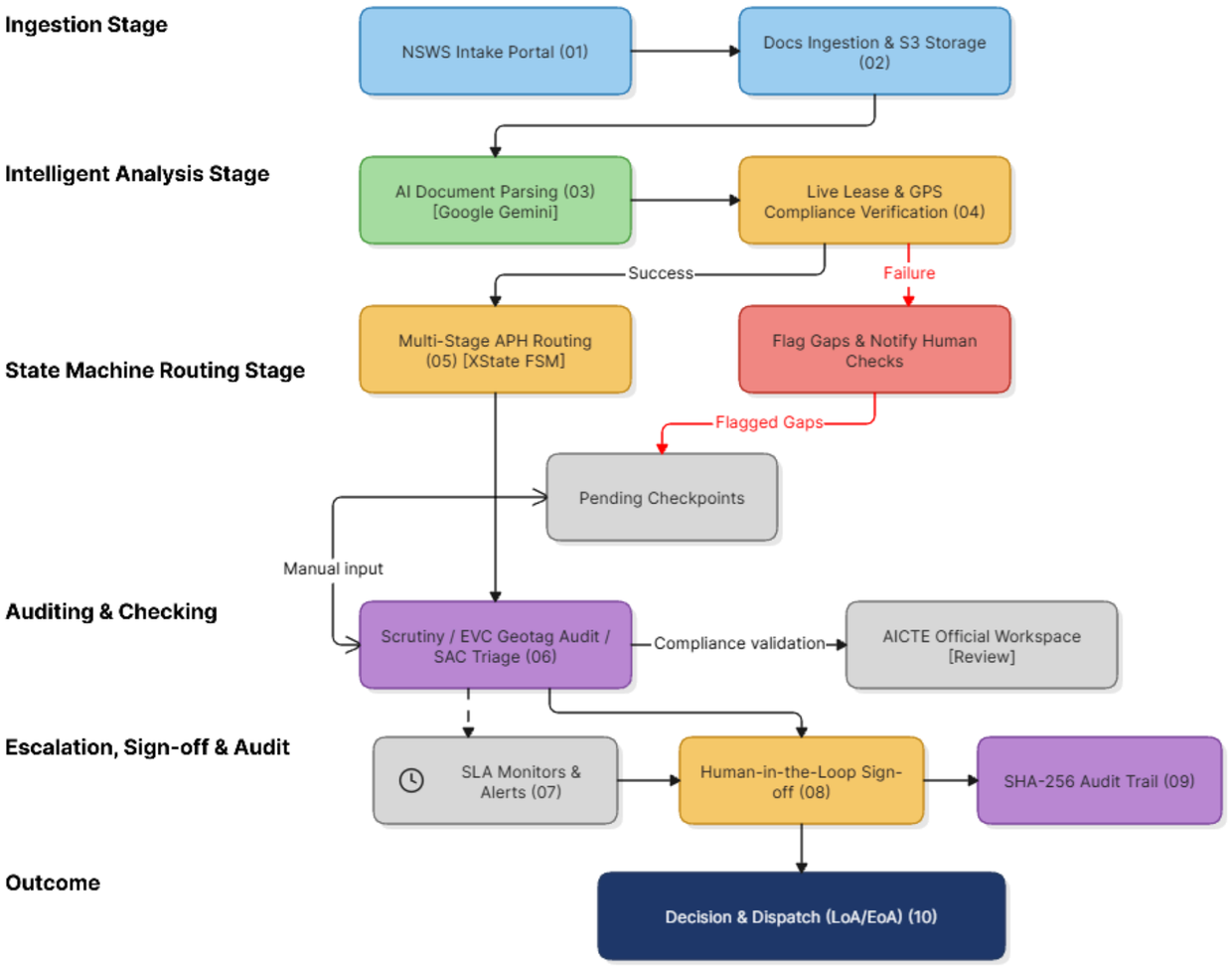
TECHNICAL APPROACH

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Technology Stack:

- **Frontend & UI:**
- Next.js 14 & React 18: Officer dashboards and application workspaces
- TypeScript & Tailwind CSS: Typed interfaces and responsive UI styling.
- **Backend API:**
- FastAPI (Python): High-performance asynchronous API web routing.
- Pydantic: Secure data validation schemas.
- **Workflow & State Management:**
- XState: FSM orchestrating handbook-compliant, multi-stage transition logic.
- **Database & Integrity:**
- PostgreSQL: Relational database for workflows, users, and state logs.
- SHA-256 Hashing: Tamper-evident ledger.
- **Task Queue & Storage:**
- Redis & Celery: Async background task queue & proactive SLA timers.
- MinIO / AWS S3: High-capacity encrypted storage for deeds, videos.
- **AI & Media Extraction:**
- Google Gemini 2.5 Flash: Multi-modal OCR parsing lease dates/NOCs.
- OpenCV: Extract metadata/geotags from inspection logs.

Process Flowchart:



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FEASIBILITY AND VIABILITY

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Technical Feasibility

- **AI Extraction:** Gemini 2.5 Flash parses unstructured lease & land deeds into strict JSON schemas.
- **Rule Engine:** Deterministic backend checks for land area, survey numbers, and live lease validity.
- **State Orchestration:** Finite State Machine controls transitions across Scrutiny, EVC, SAC, and SHC.
- **Core Tech Stack:** Built on Next.js 14, FastAPI, PostgreSQL, Celery, and Redis.
- **Dockerized Core:** Reproducible environment running APIs, background workers, and persistent stores.

Operational & Implementation Viability

- **Central Command:** Role-based workspaces with live application queues and SLA indicators.
- **Configurable SLAs:** Background monitors tracking stage-wise pendency (15-day baseline).
- **Multi-Tier Escalation:** Auto-flagging overdue files at 15, 30, and 45-day intervals.
- **Modular Deployment:** Docker Compose architecture ready for on-premise government servers or cloud.
- **Scalable Storage:** PostgreSQL indexing paired with S3/MinIO for secure, high-volume document archives.

Challenges

- **Regulatory Shifts:** APH handbook rules and approval criteria update annually.
- **Document Ambiguity:** Low-res scans or irregular deed formats degrade OCR extraction.
- **External API Barriers:** State land registries lack open, standardized verification APIs.
- **Workflow Oversight:** Automation must not obscure officer responsibility.
- **Alert Delivery Limits:** Current prototype relies on dashboard-only alerts.

Mitigation Strategy

- **Configurable Rules:** Decoupled business rules allow updates without code refactoring.
- **Human-in-the-Loop Triage:** Low-confidence parses trigger flagged manual verification checkpoints.
- **Simulated Adapters:** Standalone validation now; direct API hooks added as portals open.
- **Tamper-Evident Ledger:** Cryptographic SHA-256 audit trail logging automated and manual actions.
- **Asynchronous Hooks:** Celery/Redis queue primed for SMS/Email gateway integration.

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IMPACT AND BENEFITS

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Direct Impact on Target Users

- **AICTE Scrutiny Officers:** Reduce repetitive manual checking through AI-assisted lease-document extraction and rule-based validation.
- **EVC, SAC & SHC Teams:** Improve application-stage visibility and support consistent workflow transitions.
- **Regional Authorities:** Identify overdue applications through stage-wise SLA monitoring and dashboard escalation alerts.
- **Technical Institutions:** Receive more consistent document checks and clearer identification of incomplete or non-compliant lease information.
- **Administrative Teams:** Access centralized application records, document references, and workflow history for easier review.

Economic & Operational Benefits

- **Reduced Manual Work:** Automates parts of lease-data extraction and configured compliance checking.
- **Improved Pendency Visibility:** Highlights applications that exceed configured stage-wise time limits.
- **Better Resource Utilization:** Helps officers prioritize overdue cases and focus attention on exceptions.
- **Scalable Digital Infrastructure:** Uses modular APIs, PostgreSQL, background processing, and scalable cloud document storage.

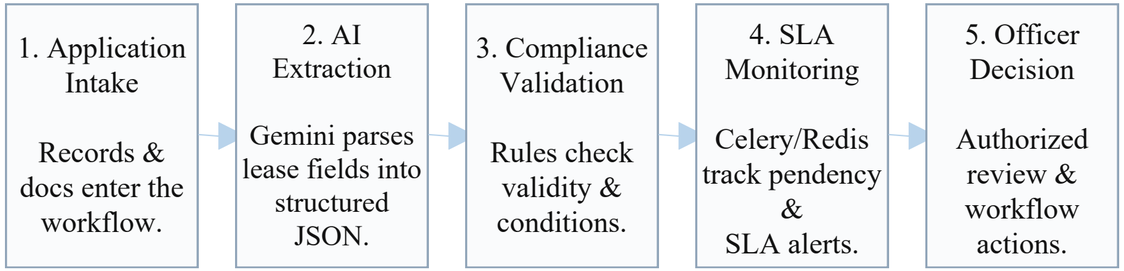
Strategic Impact

- **APH-Aware Compliance:** Supports consistent application routing and configured regulatory checks.

- **Greater Transparency:** Connects workflow actions and validation outcomes with traceable application records.
- **Accountable Decision-Making:** Preserves authorized officer oversight for approvals, rejections, and exception handling.

- **Digital Governance:** Establishes a foundation for structured, traceable, and scalable AICTE application processing.

VERIFLOW-AICTE: Workflow & Value Impact



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RESEARCH AND REFERENCES

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Gap & Problem Identification

- **Manual Verification:** Extensive manual checks for application documents create repetitive workflows. [1]
- **Incomplete Dossiers:** Missing, expired, or incorrect documents complicate scrutiny and compliance reviews. [1]
- **Exhaustive Compliance:** High manual overhead tracking multiple clauses in AICTE's approval handbook. [1]
- **SLA Tracking:** Multi-stage transitions make stage-wise pendency and overdue actions difficult to monitor.

Economic & Strategic Landscape

- **Administrative Efficiency:** Automating parts of document extraction and compliance checking reduces repetitive officer workload.
- **Process Transparency:** Centralized workflow records and stage-wise monitoring make application status easier to review.
- **Scalable Governance:** Modular APIs, database-backed records, and background monitoring provide a foundation for extending the prototype.
- **Human Accountability:** Automated checks support authorized officers rather than replacing statutory decision-making.

Literature Survey & Competitive Analysis

- **Rombach & Fettke (2025):** Systematic review examining 130 deep-learning approaches for key information extraction (KIE) from business documents. [2]
- **Baviskar et al. (2021):** IEEE Access review highlights complex layouts and dataset quality as primary automated extraction challenges. [3]
- **Castellanos Ardila et al. (2022):** Large-scale study of automated process compliance, identifying representation and traceability gaps. [4]

Field Tests & Prototype Validation

- **Lease Extraction:** Assess Gemini extraction accuracy for lease dates, land area, and survey numbers into JSON.
- **Compliance Validation:** Test lease-duration rules and handling of missing, invalid, or ambiguous fields.
- **Workflow Validation:** Check permitted transitions across Scrutiny, EVC, SAC, SHC, and final decisions.
- **SLA Monitoring:** Verify configured stage-age checks and dashboard escalation levels beyond 15, 30, and 45 days.
- **Audit Verification:** Test if SHA-256 chained audit records detect unauthorized modifications to recorded history.

Technology Benchmarking

- **AI Extraction using Google Gemini 2.5 Flash:** converting unstructured lease-document content into JSON.
- **Validation:** Python backend rules validating extracted values and configured compliance conditions.
- **Dual-Layer FSM:** Client-side XState wizards backed by a server-side Python transition engine.
- **SLA Monitoring:** Celery and Redis supporting scheduled SLA checks and dashboard escalation alerts.
- **Auditability:** SHA-256 chained audit records supporting integrity verification and review.

Policy & Ecosystem Analysis

- **Regulatory Foundation:** AICTE Approval Process Handbook (2024–2027) provides the relevant regulatory context for processing. [1]
- **Compliance Configuration:** Workflow and document rules must map to handbook provisions and application categories.
- **Authorized Verification:** External document verification requires issuing-authority access, APIs, or portal integrations.
- **Responsible Automation:** Human review, traceable actions, and clearly defined decision authority are essential.